MeetBot (AI Meeting Assistant)

Project Overview

MeetBot is an AI-powered meeting assistant that converts **speech into text and automatically generates structured meeting summaries**. It helps organizations keep track of important discussions, decisions, and action items.

Problem Statement

Important information discussed in meetings is often lost or poorly documented. Automated transcription and summarization can improve productivity.

Technologies Used

Programming Language - Python
Speech Processing - SpeechRecognition, PyAudio
Natural Language Processing - NLTK, spaCy
Text Summarization - Transformers, BERT Models
Data Processing - Pandas, NumPy
Backend Development - Flask
Data Storage - JSON, Firebase
Tools & Environment - Google Colab, GitHub

Methodology

1. Audio Capture and Input Processing

- The system captures **meeting audio input** using a microphone or uploaded audio file.
- Audio signals are recorded using the **PyAudio library**.
- The recorded audio is stored temporarily for processing.

2. Speech-to-Text Conversion

- The recorded audio is processed using the **SpeechRecognition library**.
- SpeechRecognition converts spoken language into **machine-readable text transcripts**.
- The transcription engine processes the audio frame-by-frame to improve recognition accuracy.

3. Text Preprocessing

- The generated transcript is cleaned and prepared for NLP processing.
- Preprocessing steps include:
    - **Lowercasing text**
    - **Removing punctuation and noise**
    - **Tokenization**

    - **Stop-word removal**
    - **Lemmatization or stemming**
- Libraries such as **NLTK and spaCy** are used for this stage.

4. Natural Language Processing (NLP) Analysis

- The cleaned transcript is analyzed using NLP techniques to understand the meeting content.
- Important components extracted include:
    - **Key topics discussed**
    - **Important sentences**
    - **Entities such as names, tasks, or deadlines**
- NLP helps identify the most meaningful parts of the meeting conversation.

5. Text Summarization

- A **Transformer-based model (BERT or similar)** is used for summarization.
- The model analyzes the transcript and generates a **concise meeting summary**.
- Summaries highlight:
    - Key discussion points
    - Decisions made during the meeting

6. Action Item Extraction

- NLP algorithms identify **task-related sentences** within the transcript.
- These sentences are classified as **action items**.
- The system organizes tasks into structured bullet points.

7. Data Structuring and Storage

- The generated transcript, summary, and action items are structured into a **JSON format**.
- Data can optionally be stored in **Firebase database** for future access.

8. Output Generation

- The final output includes:
    - Full meeting transcript
    - Summary of the meeting
    - List of action items and key decisions
- The results are displayed in a **structured interface for easy review**.

Key Features

- Automatic speech transcription
- Meeting summarization
- Key points extraction
- Action item generation

Results

MeetBot significantly reduces manual effort required for meeting documentation.